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Maths - Trigonometry Unit Test

Section A

Q1. (a) 4/5

Q2. (c) 1

Q3. (b) $\text{cosec}^2 A$

Q4. (b) 45°

Q5. (b) 9

Q6. (c) $\sqrt{3}$

Q7. (c) $30\sqrt{3}$ m

Q8. (c) $\sqrt{3}$

Section B

Q9. $3 \cot A = 4 \Rightarrow \cot A = 4/3$

$\tan A = 1/\cot A = 3/4$, $\tan^2 A = 9/16$

$$\frac{1 - \tan^2 A}{1 + \tan^2 A} = \frac{1 - 9/16}{1 + 9/16} = \frac{7/16}{25/16} = 7/25$$

Ans = 7/25

Q10. $\tan 45° = 1$, $\cos 30° = \sqrt{3}/2$, $\sin 60° = \sqrt{3}/2$

$2 \tan^2 45° = (2 \times 1)^2 = 4$

$= 4 + (\sqrt{3}/2)^2 - (\sqrt{3}/2)^2$

$= 4 + 3/4 - 3/4 = 4$

Q11. $\sin(A - B) = 1/2 = \sin 30°$

$A - B = 30° \ldots(1)$

$\cos(A + B) = 1/2 = \cos 60°$

$A + B = 60° \ldots(2)$

adding (1) and (2) $\Rightarrow 2A = 90° \Rightarrow A = 45°$

$B = 60° - 45° = 15°$

Q12. $LHS = (\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

$= \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta$

$= 2\sin^2\theta + 2\cos^2\theta = 2(\sin^2\theta + \cos^2\theta)$

$= 2 \times 1 = 2 = RHS$ Hence proved

Q13. $LHS = (\sec A - \tan A)^2$

$$= \left(\frac{1}{\cos A} - \frac{\sin A}{\cos A}\right)^2$$

$$= \frac{(1 - \sin A)^2}{\cos^2 A}$$

$$= \frac{(1 - \sin A)^2}{1 + \sin^2 A}$$

$$= \frac{1 - \sin A}{1 + \sin A} = RHS$$

Q14. $\sec\theta = 13/12 \Rightarrow \cos\theta = 12/13$

$\sin\theta = \sqrt{(1 - \cos^2\theta)} = \sqrt{(1 - 144/169)} = \sqrt{(25/169)} = 5/13$

$\tan\theta = \sin\theta/\cos\theta = 5/12$

$\cot\theta = 5/12$

$\operatorname{cosec}\theta = 13/5$

Q15. $$LHS = \frac{\cos A}{1 + \sin A} + \frac{1 + \sin A}{\cos A}$$

$$= \frac{\cos^2 A + (1 + \sin A)^2}{\cos A\,(1 + \sin A)}$$

$$= \frac{\cos^2 A + 1 + \sin^2 A + 2\sin A}{\cos A\,(1 + \sin A)}$$

$$= \frac{2 + 2\sin A}{\cos A\,(1 + \sin A)} = \frac{2\,(1 + \sin A)}{\cos A\,(1 + \sin A)}$$

$= 2/\cos A = 2\sec A = RHS$

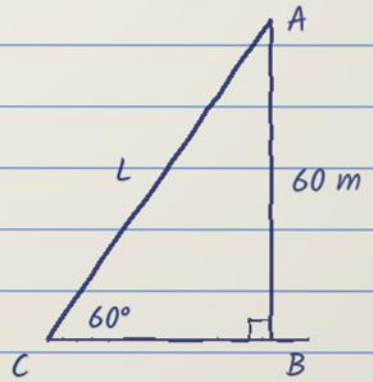
Q16.

Let length of string $AC = L$

$$\cos 60° = \frac{AB}{AC}$$

$$\frac{1}{2} = \frac{60}{L}$$

$L = 120$ m



Q17. $LHS = \dfrac{\tan\theta}{1-\cot\theta} + \dfrac{\cot\theta}{1-\tan\theta}$

$= \dfrac{\tan\theta}{1-1/\tan\theta} + \dfrac{1/\tan\theta}{1-\tan\theta}$

$= \dfrac{\tan^2\theta}{\tan\theta - 1} - \dfrac{1}{\tan\theta(\tan\theta - 1)}$

$= \dfrac{\tan^3\theta - 1}{\tan\theta(\tan\theta - 1)}$

$= \dfrac{\tan^2\theta + 1}{\tan\theta} = \dfrac{\sec^2\theta}{\tan\theta}$

$= \dfrac{1}{\sin\theta\cos\theta} = \sec\theta\,\mathrm{cosec}\,\theta = RHS$

Q18.

$AB = 7$ m, let $BD = AE = x$

In ΔABD, $\tan 45° = AB/BD$

$1 = 7/x \Rightarrow x = 7$ m

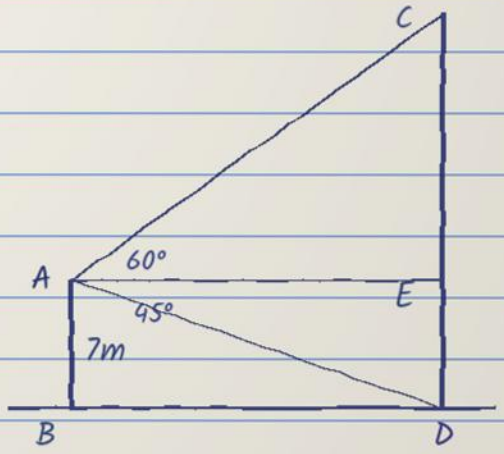
In ΔAEC, $\tan 60° = CE/AE$

$\sqrt{3} = CE/7 \Rightarrow CE = 7\sqrt{3}$ m

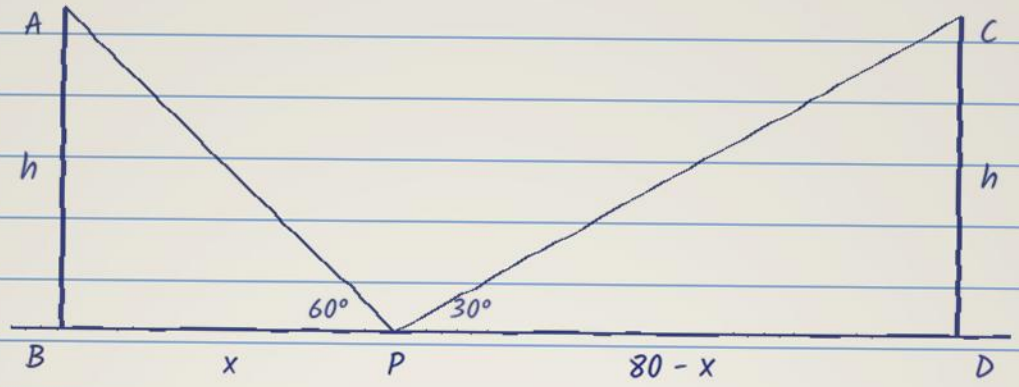
Height of tower $CD = CE + ED = 7\sqrt{3} + 7$

$= 7(\sqrt{3} + 1) = 7 \times 2.732 = 19.124$ m

Height of tower = 19.12 m (approx)



Q19.



Let BP = x , then PD = 80 - x , AB = CD = h

In ΔABP , $\tan 60° = h/x \Rightarrow h = \sqrt{3}\,x$ ...(1)

In ΔCDP , $\tan 30° = h/(80 - x) \Rightarrow h = \frac{80 - x}{\sqrt{3}}$ ...(2)

from (1) and (2) , $\sqrt{3}\,x = \frac{80 - x}{\sqrt{3}}$

$3x = 80 - x$

$2x = 80$

$x = 40$ m

$h = \sqrt{3} \times 40 = 40\sqrt{3} = 69.28$ m

Height of poles = 69.28 m

Distances of point from poles = 40 m and 40 m